Fair Test · Teacher and Cover Plan

Learning goal: I can plan a fair test by naming what to change, what to measure, and what to keep the same.

Scheme Outcome

TRACK Science scheme of work: Forces and Fair Testing

Plan a scientific enquiry to answer a force question, identifying the independent, dependent, and control variables.

Success Criteria

I can turn a science question into a testable plan.

I can name the one independent variable to change and the dependent variable to measure.

I can list at least three control variables to keep the same and explain why.

Model Sequence (I do)

First, name the independent variable (what we change): the area of the parachute canopy.

Second, name the dependent variable (what we measure): the fall time in seconds (s).

Third, lock the control variables: same drop height (1.5 m), same 10 g mass, and same strings.

A good fair-test plan also includes repeated trials (for example, 3 drops for each parachute size).

Repeating each size lets us spot a timing mistake or sudden draught and check consistency.

Always state the measurement units in your plan: cm for length, g for mass, and s for time.

We Do & Hinge Check

We want to test if ramp height affects how far a toy car rolls. Which variable is the independent variable?

Answer: A. Ramp height is what we change on purpose (IV); rolling distance is what we measure (DV).

Safe Practical & Card Setup

Sort the variable cards onto the Change / Measure / Keep the Same planning mat and write your investigation plan.

Kit: Printed Variable Sorting Cards, 3-box Fair-Test Planning Mat, sample parachute kit, and stopwatch.

Keep the floor drop zone clear. Stand on the floor only (never stand on chairs or desks to drop a parachute).

You Do Routes (Supported, Independent, Aspire)

Supported: Match Canopy Size, Fall Time (s), and Drop Height (1.5 m) to Change, Measure, and Keep the Same. (Help: Change = canopy size. Measure = fall time in seconds. Keep the same = drop height and load mass.)

Independent: Complete the fair-test plan for the parachute investigation (IV, DV with units, 3 controls) and write a scientific prediction explaining why a larger canopy should fall more slowly. (Help: We will change __, measure __ in __, and keep __, __, and __ the same. I predict the largest canopy will take

the longest time because __.)

Aspire: Explain why we plan 3 repeat drops for each canopy size and how a human reaction-time delay on the stopwatch could affect the results. (Help: Three repeats let us spot an __ and calculate a reliable middle/mean value. Reaction time adds a small uncertainty to __.)